

Honest Invalidation of a Cross-Sectional Equity Signal: A López de Prado Pipeline on the Brazilian Stock Market

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Abstract

We apply a deliberately rigorous Advances-in-Financial-Machine-Learning (AFML) pipeline to the cross-sectional prediction of trade success on 111 liquid Brazilian (B3) equities over 2010–2026, and report a study of *honest invalidation*. Using triple-barrier labels, Combinatorial Purged Cross-Validation (CPCV) with purge and embargo, sample-uniqueness weighting, fractional differentiation, and a Deflated Sharpe Ratio (DSR) that deflates by a registered log of every research trial, we find four results. First, a price-derived signal (momentum, volatility, fractionally-differenced price, Amihud illiquidity, and a CUSUM/SADF-style regime block) is *statistically* real (gross DSR ≈ 0.41) but *economically* sub-marginal: net of square-root market-impact costs it collapses to a Sharpe of -1.1 when rebalanced every ten days. Second, the binding constraint is turnover $\times$ cost, not signal strength: holding the ruler fixed, the same family flips from a viable to a destroyed verdict purely as a function of rebalance frequency. Third, a simple linear value/quality fundamental tilt, rebalanced quarterly and pre-registered before estimation, is the only net-positive configuration we find (net Sharpe $+0.18$, PSR 0.92 , DSR 0.72 deflated over 18 trials; ≈ 0.35 annualized). We stress that DSR 0.72 sits *below* the conventional 0.95 skill-confidence threshold: the tilt clears its pre-registered net-positive criterion and is the best surviving configuration, but the evidence *leans toward* rather than *confirms* genuine skill, and the annualized Sharpe is modest. Fourth, state-of-the-art machine learning does not beat the linear composite: a LightGBM LambdaRank ranker overfits (out-of-sample Rank IC -0.014), and the TabPFN v2 tabular foundation model merely *ties* a zero-parameter three-factor linear composite on the fundamental factors and *loses* on the full feature set. The bottleneck is information and transaction cost, not model capacity. We frame the contribution as a reproducible anti-self-deception methodology together with an honest negative result.

1. Introduction

The efficient-market hypothesis (Fama 1970) sets a high bar for any claim of predictable, tradeable structure in security prices: a candidate signal must survive not only statistical scrutiny but also the friction of execution. In practice, the dominant failure mode of quantitative equity research is not the absence of in-sample structure but its *spurious abundance*. With enough features, enough model families, and enough rebalancing rules, a backtest can be tuned to exhibit an arbitrarily attractive Sharpe ratio that is pure selection bias. Harvey, Liu and Zhu (Harvey, Liu, and Zhu 2016) document hundreds of published “factors,” most of which fail to clear a multiple-testing hurdle. Bailey and López de Prado (Bailey and López de Prado 2014) and Bailey et al. (Bailey et al. 2017) formalize the mechanism: when a researcher selects the best of many trials, the maximum observed Sharpe ratio is inflated by an amount that grows with the number and variance of the trials, so that a nominally significant result is, in expectation, a statistical fluke.

The problem is compounded in three ways that are especially acute for an emerging market such as Brazil. (i) *Transaction cost*: cross-sectional long-short signals are turnover-intensive, and on mid-cap B3 names the price impact of trading is convex and large relative to the thin edge of any price-derived predictor (Amihud 2002; Tóth et al. 2011). (ii) *Overlapping labels*: holding-period labels sampled daily induce strong serial dependence, inflating t -statistics by roughly $\sqrt{H}$ for a horizon of H days and corrupting the very ruler used to adjudicate skill (López de Prado 2018). (iii) *Survivorship and corporate-action integrity*: a universe defined by today’s liquid index, or prices not adjusted for total return, biases base rates upward.

This paper does not present a winning strategy. Instead it presents a *methodology of honest invalidation* and the negative result it produces. We build a pipeline following Advances in Financial Machine Learning (López de Prado 2018) in which every component is chosen to *remove* a self-deception channel rather than to add edge: triple-barrier labels matched to the purge horizon, Combinatorial Purged Cross-Validation with embargo, sample-uniqueness weighting, causal fractional differentiation, and a Deflated Sharpe Ratio deflated by a registered log of every trial we ran. Within this ruler we report four findings (Section 4): a price-derived signal that is statistically real but economically sub-marginal; a demonstration that turnover $\times$ cost, not signal strength, is the binding constraint; a single pre-registered fundamental tilt that is the only viable configuration; and the result that state-of-the-art machine learning does not beat a zero-parameter linear composite. The contribution is the reproducible ruler plus the honest verdict it returns.

2. Data

Universe. We study 111 liquid Brazilian equities listed on B3, selected by an average daily traded value of at least R\$3,million and a minimum of 300 daily bars of history. The sample spans 2010-01-01 to 2026-06-15. The universe deliberately includes initial public offerings after 2018, which partially mitigates survivorship bias relative to the common practice of taking only names that are liquid *today*; a fully point-in-time index-membership reconstruction remains future work, and we treat residual survivorship as an *optimistic* bias on all reported numbers.

Prices. Daily total-return series are obtained by REST scraping of the public brapi service, using the `adjustedClose` field, which folds dividends, interest on own capital (JCP), splits and bonuses into a single back-adjusted series. The authoritative public source for raw Brazilian prices is the exchange’s COTAHIST files, which are *unadjusted*; the principal difficulty in constructing Brazilian panels is not price availability but the discontinuity of asset identity across corporate reorganizations (ticker and trading-name changes), which fragments a company’s dividend history across multiple names. Reconstructed series were validated against an independent reference (e.g., ABEV3 reconstructed from COTAHIST plus the exchange dividend feed matches within $\approx 2\%$, the residual attributable to JCP and convention differences).

Fundamentals. Value and quality factors (earnings yield, book-to-price, profit margin) are taken from quarterly statements and lagged by 90 days to respect the Brazilian regulatory (CVM) filing deadline, then combined with the *live* (non-lagged) price so that the resulting ratios are point-in-time correct: the numerator is known, the denominator is contemporaneous.

3. Methodology

Every methodological choice below is included because it closes a specific channel through which a backtest can lie. We state, for each, *what* it does and *why* it is necessary.

3.1 Triple-barrier labels

Following AFML chapter 3 (López de Prado 2018), each observation is labeled by the first of three barriers to be touched within a fixed horizon: an upper profit-taking barrier and a lower stop-loss barrier, both set at $2\sigma_t$ where σ_t is an exponentially-weighted daily volatility (span 20), and a vertical barrier at $H = 10$ trading days. The label is the sign of the realized event, and t_1 is the *event-resolution date* (the touch), not the vertical barrier. *Why*: labeling by the barrier actually hit, and propagating its resolution date t_1 into the validation purge, makes the predicted quantity, the realized return, and the leakage control mutually consistent. Labeling on a fixed horizon while purging on a different one measures one signal and validates another.

3.2 Combinatorial Purged Cross-Validation with embargo

We use CPCV (AFML chapter 12) with $N = 6$ groups and $k = 2$ test groups per split, giving $\binom{6}{2} = 15$ splits and 5 reconstructed backtest paths. Two leakage controls are applied at every train/test boundary. *Purge*: training observations whose label-resolution interval $[t_0, t_1]$ overlaps the test set are removed. *Embargo*: a fixed window after each test block is excluded from training. Critically, the embargo is set to 55 trading days, *not* the 10-day label horizon, because the embargo must cover both information horizons present in the panel: the label resolution ($H = 10$) and the memory of the fractionally-differenced feature (≈ 55 bars; Section 3.4). Without the second term, test-period prices leak into training features through the FFD convolution, a subtle indirect leak that an adversarial audit of the pipeline identified. *Why CPCV at all*: a single train/test split yields one Sharpe estimate with no distribution; CPCV yields a distribution of out-of-sample paths from which the dispersion of the performance estimator can be measured, while purge and embargo keep each path causal.

3.3 Deflated and Probabilistic Sharpe Ratio with a trials log

We evaluate economic significance with the Probabilistic Sharpe Ratio (PSR) and Deflated Sharpe Ratio (DSR) of Bailey and López de Prado (Bailey and López de Prado 2014). Two corrections are essential and are easy to get wrong.

Effective sample size. The PSR/DSR confidence in a Sharpe estimate scales with $\sqrt{n - 1}$. Naively concatenating the test returns of all CPCV splits multiplies n several-fold (each date appears in multiple test combinations) and, on top of that, ten-day overlapping returns are serially dependent. We therefore deduplicate by (ticker, date) and set the effective n to the number of unique dates divided by the horizon and further scaled by average sample uniqueness, so that the ruler counts independent bets, not redundant ones.

Deflation by real research effort. The DSR deflates the observed Sharpe by the expected maximum of the trials searched. A common error is to treat the variance of the CPCV paths as the variance of *research trials* and to set the trial count to the number of splits. CPCV paths are resamplings of a *single* configuration (estimator dispersion), not independent trials of a multiple-testing search. We instead maintain an explicit append-only **log of every research trial** — every lever, feature family, and hyperparameter configuration we ran, 21 by the end of the study — and deflate each

result by the dispersion and count of *those* Sharpe ratios *as logged at the time that result was computed* (the fundamental tilt of Section 4.3 is the 18th trial and is therefore deflated by 18; the later machine-learning runs of Section 4.4 are deflated by 20–21). *Why*: this is the only defense against the garden of forking paths. A DSR that deflates by an honest trial count is conservative by construction; one that deflates by a cross-validation knob is “right for the wrong reason” and would fail to detect skill if it existed.

3.4 Fractional differentiation

Price levels are non-stationary; returns are stationary but memoryless. Following AFML chapter 5 (López de Prado 2018) we apply fixed-width-window fractional differentiation (FFD) with order $d = 0.4$ and a weight-truncation threshold of 10^{-3} , which yields a causal window of ≈ 55 bars. *Why*: $d = 0.4$ is the smallest differencing order that renders the price series stationary while preserving the maximum amount of memory, so the resulting feature retains predictive structure that a first difference (return) discards. In our price-derived ablation this single feature accounts for the large majority of the realized discrimination gain.

3.5 Sample-uniqueness weighting

Overlapping labels make daily samples redundant: when a ten-day label is sampled every day, consecutive observations describe nearly the same outcome. Following AFML chapter 4 (López de Prado 2018) we compute the average uniqueness of each label from the concurrency of live t_1 intervals and pass it as a sample weight to every estimator. *Why, and a deliberate restriction*: we weight by uniqueness *only*, not by uniqueness $\times$ return attribution. Uniqueness down-weights redundant, crowded regimes and *preserves* the edge; return-attribution weighting, by contrast, up-weights large-move samples and we found it *destroys* the measured edge, conflating outcome magnitude with sample importance. Uniqueness also rescales the effective n used in the PSR/DSR (Section 3.3).

3.6 Transaction-cost model: square-root law, one-way

Gross long-short Sharpe ratios are optimistic fiction by construction. We charge a cost composed of a linear spread/brokerage term plus a market-impact term following the empirical square-root law of Tóth et al. (Tóth et al. 2011),

$$\text{impact} \approx \eta \sigma \sqrt{Q/\text{ADV}},$$

where Q is order size, ADV is average daily traded value in R\$, and σ is daily volatility. The square-root (concave) form is more realistic than the linear impact of Almgren-Chriss, which we treat only as historical context for scheduled execution rather than as an aggregate cost model. Cost is charged per *one-way* leg proportional to period turnover, *not* per round trip, because turnover already accounts for each side of a rotation as it occurs over time. We treat the chosen half-spread (5–10 bps) and $\eta = 1$ as a *conservative floor*: on low-ADV B3 names the true impact is more convex than the square root and the spread widens under liquidity stress, so the reported costs *underestimate* the tail. *Why this matters*: as Section 4 shows, the cost model is not a footnote; it is the single component that reverses the verdict on the price-derived family.

All features are computed point-in-time; the causal chain was audited adversarially for leakage, and a suite of 26 `pytest` regression tests locks the label $\leftrightarrow t_1 \leftrightarrow$ return alignment, the purge/embargo geometry, the FFD weights, and the PSR/DSR formulas.

4. Results

We report four findings. All economic numbers are net of the square-root, one-way cost model (Section 3.6) unless explicitly labeled “gross,” and all significance is deflated by the registered trials log (Section 3.3). The numbers are reported as measured; none are inflated or rounded in the favorable direction.

4.1 The price-derived signal is statistically real but economically sub-marginal

The price-derived family stacks momentum and volatility features, the fractionally-differenced price (Section 3.4), an Amihud illiquidity microstructure block (Amihud 2002), and a regime block built from CUSUM and an SADF-style explosiveness statistic (Phillips, Shi, and Yu 2015). Table 1 traces the discrimination of this family as components are added. Cross-sectional AUC rises from a near-coin-flip baseline of 0.5125 to 0.5232, with fractional differentiation contributing roughly three-quarters of the gain — a *real* improvement in ranking discrimination.

Table 1: Price-derived ablation. AUC is cross-sectional ranking discrimination; the economic verdict is net of costs and deflated. Discrimination improves and is attributable mostly to fractional differentiation, yet the economic edge never separates from zero.

Lever	Component added	AUC	Δ AUC	Economic verdict
L0	baseline (returns)	0.5125	—	none (single split)
L1	clean total-return panel	0.5125	+0.0000	de-risks data, no edge
L2	CPCV + DSR ruler	0.5146	+0.0021*	reveals no economic edge
L3	fractional differentiation	0.5232	+0.0086	only real AUC jump
L4	meta-labeling	0.5224	−0.0008	first negative marginal

* L2 changed the ruler, not the feature; its Δ AUC is not comparable.

Measured as a strategy, the full price-derived family achieves a *gross* DSR of ≈ 0.41 . Net of the square-root cost model, rebalanced every ten days, it collapses to a Sharpe of -1.1 — sub-economic. The result is robust: decile construction does not rescue it, a longer horizon does not rescue it, and a spread-only floor leaves it near zero even on the largest, most liquid names. The discrimination is genuine; the conversion to profit-and-loss is destroyed by trading friction.

4.2 The binding constraint is turnover $\times$ cost, not signal strength

Holding the ruler, the universe, and the signal fixed, the economic verdict is governed by *turnover*. Table 2 contrasts the same price-derived family at high and low rebalance intensity. A high-turnover configuration with a barely-positive gross Sharpe of $+0.06$ is driven to a net Sharpe of -1.1 ; the cost wedge, not the signal, decides the outcome. This is the central diagnostic of the paper: a fixed ruler returns *opposite* verdicts for high- versus low-rotation deployments of an identical predictor.

Table 2: Turnover contrast for the price-derived family. The signal is unchanged; only rebalance frequency differs. Cost, scaling with turnover, reverses the verdict.

Configuration	Rebalance	Gross Sharpe	Net Sharpe	Verdict
price, high-turnover	10 days	+0.06	−1.1	destroyed by cost
price, spread-only floor	10 days	≈ 0	≈ 0	no edge to spend

4.3 A pre-registered fundamental tilt is the only viable configuration

We pre-registered, before estimation, a single hypothesis (H1): a long-short value/quality tilt, rebalanced *quarterly* (63 trading days), would clear a net Sharpe above zero after one-way costs, precisely because low turnover keeps the toll beneath a weak-but-real fundamental edge. The construction is fixed and untuned: the score is the equal-weighted average of the cross-sectional percentile ranks of earnings yield, book-to-price and profit margin (Fama and French 1992, 1993; Novy-Marx 2013); the portfolio is long the top tercile, short the bottom tercile, equal-weighted and market-neutral; the return is the forward return of each name to the next rebalance; the acceptance criterion was net Sharpe > 0 , with the complementary outcome declared a confirmation of the kill-criterion.

Table 3: Pre-registered quarterly fundamental tilt. The first and only economically viable configuration in the study. DSR is deflated over the full 18-trial log; the annualized figure scales the per-rebalance net Sharpe by $\sqrt{252/63}$.

Quantity	Value
Sharpe, gross	+0.19
Sharpe, net (one-way costs)	+0.18
PSR ($H_0 : \text{SR} = 0$)	0.92
DSR (deflated, 18 trials)	0.72
Sharpe, net, annualized	≈ 0.35

The pre-registered hypothesis passed its stated criterion (net Sharpe > 0): net Sharpe +0.18, PSR 0.92, and a DSR of 0.72 deflated over all 18 logged trials. We are careful not to overstate this. PSR 0.92 and DSR 0.72 both fall *short* of the conventional 0.95 confidence threshold for declaring genuine skill; the result clears the (weaker) pre-registered net-positive bar and is the only net-positive configuration in the study after honest costs and honest deflation, but it is best read as *promising, not confirmed*, and the annualized Sharpe (≈ 0.35) is modest. Its (tentative) viability is *mechanistic*: quarterly rebalancing cuts turnover by roughly an order of magnitude relative to the ten-day price-derived deployment, moving the same class of weak edge from the wrong to the right side of the cost wedge identified in Section 4.2.

4.4 State-of-the-art machine learning does not beat the linear composite

If the bottleneck were model capacity, a more expressive learner should extract more edge from the same features. It does not. Table 3 compares two state-of-the-art models against the zero-parameter three-factor linear composite of Section 4.3.

A LightGBM LambdaRank cross-sectional ranker (Ke et al. 2017; Burges 2010) over 40 features overfits: its out-of-sample Rank IC is -0.014 and its net edge is ≈ 0 , on an effective sample of only ≈ 58 quarters. The TabPFN v2 tabular foundation model (Hollmann et al. 2025), purpose-built for the small- N regime that exactly characterizes this problem, merely *ties* the linear composite on the fundamental factors ($+0.183$ versus $+0.18$) and *loses* on the full feature set, where it is pushed to a turnover of 1.10 per rebalance and churns away its edge.

Table 4: Model comparison. The foundation model ties a zero-parameter linear composite on the viable factors and underperforms on the full set. Capacity is not the constraint.

Model	Feature set	OOS Rank IC	Net edge	Note
Linear composite (3 factors)	value/quality	—	$+0.18$	zero parameters
LightGBM LambdaRank	40 features	-0.014	≈ 0	overfits, eff. $N \approx 58$
TabPFN v2	value/quality	—	$+0.183$	ties the linear composite
TabPFN v2	full features	—	$< +0.18$	turnover 1.10, churn

The verdict is unambiguous: a three-factor linear composite with no free parameters equals the SOTA tabular foundation model. The constraint is *information and cost*, not model sophistication.

5. Discussion

Turnover is the gargoyle on the gate. The four results compose into one claim: across deployments of the same family, the variable that flips the economic verdict is turnover $\times$ cost, not the strength of the predictor (Sections 4.1–4.2). A gross DSR of 0.41 is not “alpha”; it is a discrimination statistic that the square-root impact law erases before it becomes P&L. Any study that reports gross Sharpe ratios for a turnover-intensive cross-sectional signal in an emerging market is, in effect, reporting the cost it chose not to charge.

Information dominates model. That a zero-parameter linear composite ties a tabular foundation model (Section 4.4) is not a statement that TabPFN is weak — on the contrary, it is exactly the right tool for a small- N tabular problem (Hollmann et al. 2025). It is a statement about *where the edge lives*. With ≈ 58 effective quarters and three weakly-informative factors, the achievable edge is pinned by the information content of the inputs and the cost of acting on them, and no

amount of capacity manufactures signal that the data do not contain. The implication for practitioners is to spend the next unit of effort on *orthogonal, non-price information* and on *lowering turnover*, not on a larger model.

The fundamental tilt as the only net-positive configuration. The single configuration that clears the pre-registered net-positive bar is the simplest one: a slow, linear, value/quality tilt (Section 4.3). Its survival is not a triumph of cleverness but of *parsimony under friction* — a low-turnover deployment of a weak-but-real factor premium that has survived decades of out-of-sample scrutiny in developed markets (Fama and French 1992; Novy-Marx 2013). Even so, an annualized net Sharpe of ≈ 0.35 is modest, and we report it as such.

On the honesty of the deflated DSR. The DSR of 0.72 for the fundamental tilt is meaningful precisely because it is deflated by all 18 logged trials, not by a cross-validation knob (Section 3.3). The honest version of this ruler is *adversarial to its own author*: it raises the bar in proportion to how hard we searched. The price-derived family failed that bar; the fundamental tilt cleared it. We caution that our trials were sequential and adaptive, so even a registered-log deflation likely *understates* the true search effort (Bailey et al. 2017), and that residual survivorship biases all numbers optimistically.

Limitations. The universe is not yet fully point-in-time; the cost model is a conservative floor whose tail is, if anything, worse on illiquid names; the effective sample for quarterly tests is small; and the embargo and uniqueness corrections, while necessary, cannot fully repair the limited number of independent bets available in sixteen years of Brazilian data.

6. Conclusion

A rigorous López de Prado pipeline applied to Brazilian equities yields a study of honest invalidation. Price-derived signals are statistically real but economically sub-marginal once square-root transaction costs are charged; the binding constraint is turnover $\times$ cost rather than signal strength; a simple, pre-registered, quarterly value/quality fundamental tilt is the only economically viable configuration we found; and a state-of-the-art tabular foundation model does no better than a zero-parameter linear composite, identifying information and cost — not model capacity — as the bottleneck. The contribution is a reproducible, anti-self-deception methodology, and an honest negative result that we believe is more useful to the field than another un-costed, un-deflated backtest. The full pipeline, trials log, and tests are released to make the verdict reproducible.

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